

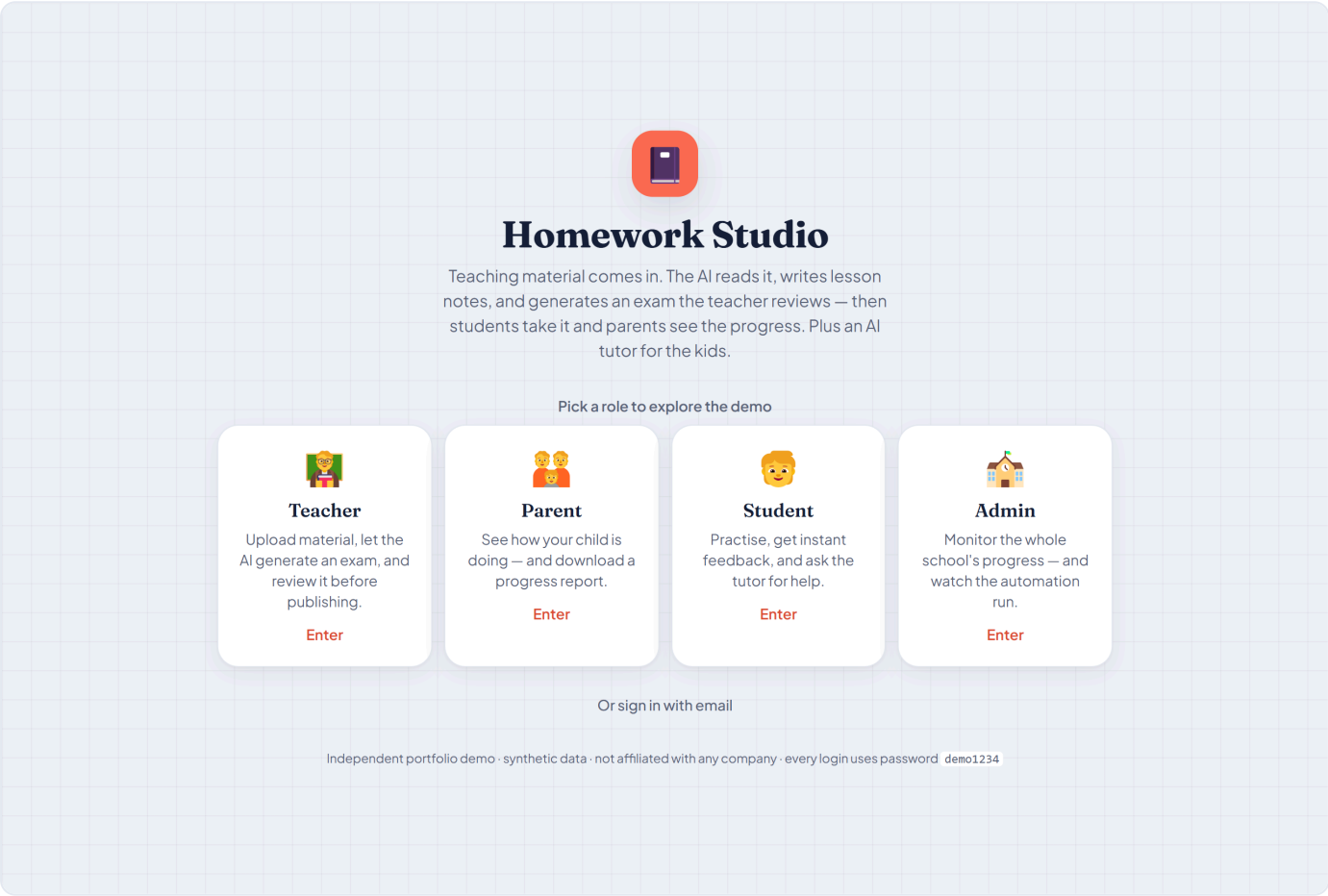
# Homework Studio — User Manual

A step-by-step guide to the four roles: Teacher, Parent, Student, and Admin.

Independent portfolio demo. All students and data are synthetic. This manual walks through what each person does, with screenshots of the live app.

## Getting started

Open the app and you land on the sign-in screen. Instead of typing a password, you pick a **role** to explore the demo — one tap signs you in as a sample user.



**Demo accounts** — every account uses the password `demo1234` :

| Role                                                                                        | Email                        | Lands on                                |
|---------------------------------------------------------------------------------------------|------------------------------|-----------------------------------------|
|  Teacher | <code>teacher@demo.id</code> | Class dashboard                         |
|  Parent  | <code>parent@demo.id</code>  | Your children's progress                |
|  Student | <code>student@demo.id</code> | Your learning home (XP, streak, badges) |
|  Admin   | <code>admin@demo.id</code>   | School overview                         |

Switch roles anytime with **Switch role** in the top-right corner.

## What you can upload — and what the AI makes from it

The teacher uploads **teaching material** (a syllabus or lesson). Nothing else in the app requires an upload.

**Accepted formats:**

| Type     | Extensions          | Read by                                                         |
|----------|---------------------|-----------------------------------------------------------------|
| Document | .pdf                | The vision model (the first page is rasterized and transcribed) |
| Image    | .png , .jpg , .jpeg | The vision model (transcribed to text)                          |
| Text     | .txt , .md          | Read directly                                                   |

- **One file per upload**, up to **25 MB**.
- `.docx` is not supported yet — export to PDF or paste into a `.txt` / `.md` first.
- On the free demo (no AI key), plain-text `.txt` / `.md` works best, since reading a PDF or photo needs the vision model.

**From one material, the AI produces three things:**

1. **A custom exam** — multiple-choice questions grounded in your material, that students take and get auto-graded.
2. **Teaching notes** — a short lesson summary (key points + a worked example).
3. **Tutor knowledge** — the material is indexed so the student tutor answers from *your* class's content.

## Teacher

**What you do:** upload material, let the AI turn it into an exam, review the questions, and publish. Then watch how the class does.

### The dashboard

You land on the class at a glance: how many students, how many exams have been taken, the class average, and how many AI questions are **awaiting your review**. Below are the four-step flow, the score distribution, the per-subject averages, and your recent materials.

Dashboard

Materials

Exams

Ms. Rida

Viewing as Teacher

Switch role

Class dashboard

Turn your material into assessments — and see how the class is doing.

Upload material

1 AI question needs your review

Approve or edit them before students take the exam.

Review now

6 Students

29 Exams taken

66% Class average

1 Awaiting review

How a lesson becomes an exam

1 Upload

Drop in a syllabus or lesson.

2 Read

The AI reads it and writes teaching notes.

3 Generate

It drafts an exam grounded in your material.

4 Publish

Review flagged questions, then publish.

Score distribution

Average by subject

Math

69%

Science

62%

Recent materials

Upload another

Fractions for Grade 4

Math · 2026-09-20

Ready

Exam →

Math Basics

Math · 2026-09-20

Ready

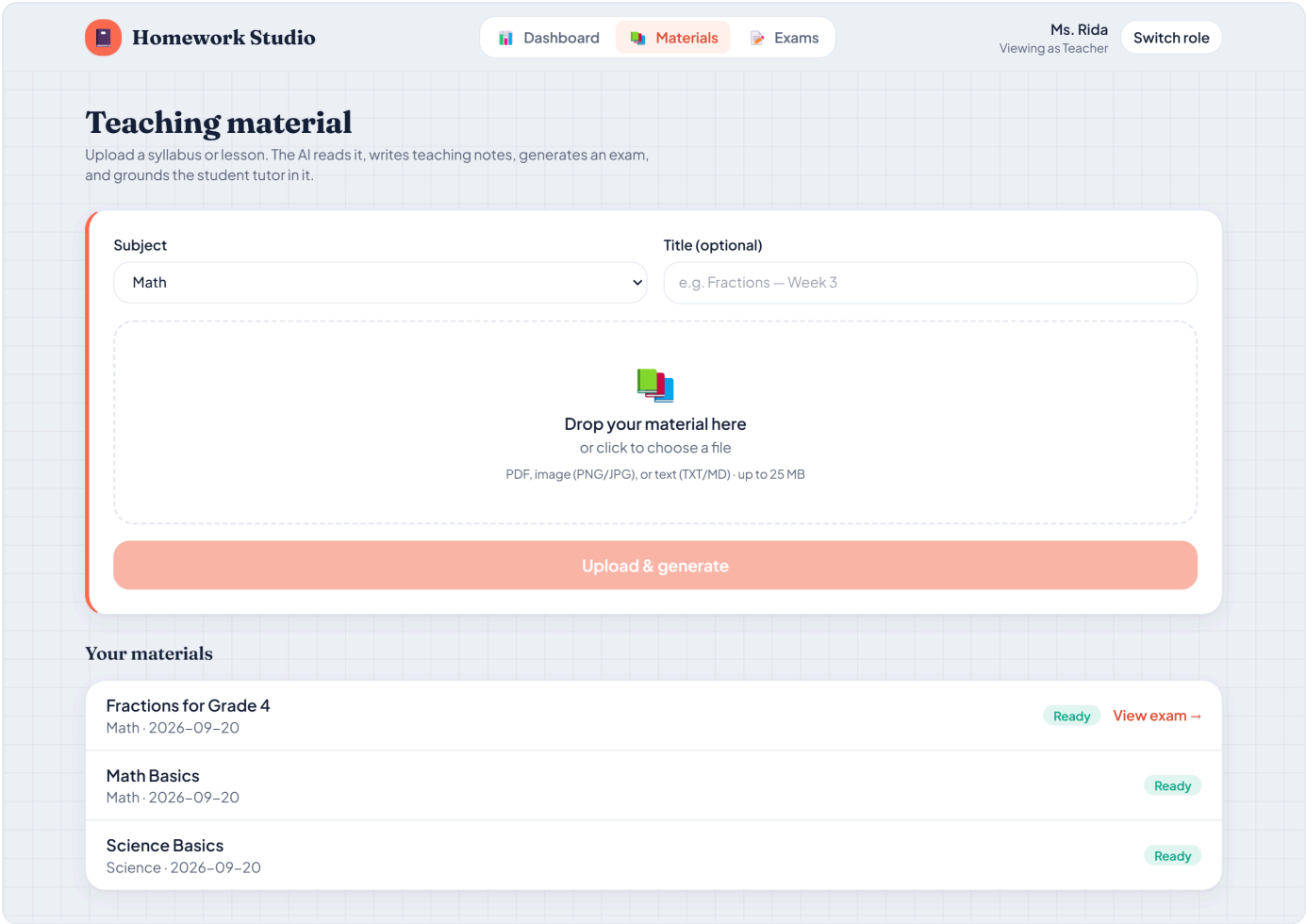
Science Basics

Science · 2026-09-20

Ready

Step by step: material → a published exam

1. Open Materials → choose a subject and (optionally) a title. 2. Drop your file onto the box, or click to browse. The accepted formats and the 25 MB limit are shown right under the drop zone.



3. Press **"Upload & generate"**. The material now moves through the pipeline. You'll see its status change:

| Status        | What it means                                                                                                                             |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Processing... | The AI is reading the material, writing notes, and generating the exam. Usually a few seconds (longer for a PDF/photo).                   |
| Ready         | Done — teaching notes were written and an exam was generated. A <b>View exam</b> → link appears.                                          |
| Failed        | The file couldn't be read (corrupt file, or an image/PDF with no AI key). Try a <code>.txt</code> / <code>.md</code> , or check the file. |

4. **Review the generated exam.** Open the exam (from **View exam** → or the **Exams** tab). Every question shows the AI's **confidence**. The ones the model was *least* sure about are **flagged for review** (amber) — that's the human-in-the-loop gate: nothing reaches students until you approve it.

[Dashboard](#)
[Materials](#)
[Exams](#)

Ms. Rida  
Viewing as Teacher
[Switch role](#)

[All exams](#)

## Fractions for Grade 4 — Exam

Review the AI-generated questions. Flagged ones are the model's least-confident reads — check them, drop any that are wrong, then publish.

[✓ Approve & publish](#)

1 question is flagged as low-confidence — worth a look before publishing.

1. Which is closest to one whole?

review - 57%

3/4 ✓

1/4

1/2

2/8

$\frac{3}{4}$  is nearest to 1.

Discard this question

2. Which fraction is bigger: 1/2 or 1/4?

✓ 94%

1/2 ✓

1/4

They are equal

You cannot tell

Halves are bigger than quarters.

Discard this question

3. What is 1/2 of 8?

✓ 92%

2

3

4 ✓

5

$8 \div 2 = 4.$

Discard this question

4. Half of a pizza is the same as \_\_\_ of the pizza.

✓ 86%

1/2 ✓

2/2

1/4

1/3

Half means  $\frac{1}{2}$ .

Discard this question

For each question you can:

- **Keep it** — the correct answer is highlighted in green; a confidence badge shows how sure the AI was.
- **Discard this question** — drops a question that's wrong or off-topic.

**5. Press "Approve & publish".** The exam's status becomes **Published** and it appears for students. That's it — the material is now an assessment.

**What "success" looks like:** the material reads **Ready**, the exam has sensible questions (flagged ones checked), and after publishing its badge is **Published**. **What "needs a look" looks like:** an exam badge of **Needs review**

with a flagged count — one or more questions the AI wasn't confident about, waiting for you.

**Exam states:**

| Exam status         | Meaning                                                                   |
|---------------------|---------------------------------------------------------------------------|
| <b>Draft</b>        | Generated; every question cleared the confidence bar. Publish when ready. |
| <b>Needs review</b> | One or more questions are flagged — check them before publishing.         |
| <b>Published</b>    | Live for students.                                                        |

 **Student**

**What you do:** take exams, practise, learn from your mistakes, and level up.

**1. Your learning home**

**Home** is a gamified overview: your **level ring** and **XP**, a **day streak**, and quick chips (average, best score). It shows your **badges** (earned + still-locked with a hint), your **subject mastery**, your **recent results**, and the **class leaderboard**. Earn XP by taking exams and practising.

Homework Studio

Home

Exams

Practice

Tutor

Aisha

Viewing as Student

Switch role

Hi, Aisha! 🙌

Your learning at a glance — level up by taking exams and practising.

Lv 3

38/120 XP

278 XP

82 XP to level 4

🔥 5-day streak

🎯 74% avg

🏆 best 100%

Take an exam

Practise

🏆 Class leaderboard

|               |        |
|---------------|--------|
| 🏆 Aisha (you) | 278 XP |
| 🏆 Chandra     | 168 XP |
| 🏆 Eka         | 164 XP |
| 4 Budi        | 160 XP |
| 5 Dewi        | 160 XP |

Badges

6 of 8 earned

🎉

First steps

Earned!

🏆

Perfect!

Earned!

🎯

Sharpshooter

Earned!

📚

Subject master

Earned!

🔍

Explorer

Earned!

🔥

On fire

Earned!

⚡

Unstoppable

Practise 7 days in a row

100

Century club

Complete 10 quizzes

Your subjects

📖 Math

100%

📖 Science

55%

Recent results

See all →

|                                          |      |
|------------------------------------------|------|
| Science Basics — Quiz                    | 20%  |
| Science Basics — Quiz                    | 20%  |
| Practice · Science <span>practice</span> | 100% |
| Math Basics — Quiz                       | 100% |

2. Take an exam

**Exams** lists every exam your teacher published. Pick one, press **Start exam**, answer each question, then **Submit** for an instant score. Tap 💡 **Show me how** anytime for a step-by-step explanation (math renders properly).

Homework Studio — independent portfolio demo

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 Homework Studio

[Home](#) [Exams](#) [Practice](#) [Tutor](#)

Aisha  
Viewing as Student [Switch role](#)

## Science Basics — Quiz

Answer each question, then submit to see how you did.

[← All exams](#)

1. Which gas do plants absorb during photosynthesis?

Oxygen

Nitrogen

Carbon dioxide

Hydrogen

 [Show me how](#)

2. What is the closest planet to the Sun?

Venus

Mercury

Earth

Mars

 [Show me how](#)

3. What organ pumps blood around the body?

Lungs

Brain

Heart

Liver

 [Show me how](#)

4. What force pulls objects toward Earth?

Friction

Magnetism

Gravity

Tension

 [Show me how](#)

5. How many legs does an insect have?

4

6

8

10

 [Show me how](#)

Submit (0/5 answered)

### 3. Practise (drill mode)

**Practice** lets you drill from the question bank by **subject** and **level** — untimed and low-stakes. It **earns XP and keeps your streak**, but is **not counted in your grades**, so you can learn without pressure.

 Homework Studio

Home

Exams

Practice

Tutor

Aisha  
Viewing as Student

Switch role

### Practice

Low-stakes — this earns XP and keeps your streak, but isn't counted in your grades.

← Back

1.  $7 + 8 = ?$

13

14

15

16

Show me how

2. What is  $\frac{1}{2}$  of 18?

6

8

9

12

Show me how

3.  $9 \times 7 = ?$

56

63

72

81

Show me how

4.  $6 \times 4 = ?$

20

22

24

28

Show me how

5.  $36 \div 6 = ?$

5

6

7

8

Show me how

Submit (0/5 answered)

#### 4. Review your results

From **Home** → **See all** (or a recent result), open any past attempt to see **every question, your answer vs the correct one, and the explanation** — so you learn from what you missed.

 **Homework Studio**

[Home](#) [Exams](#) [Practice](#) [Tutor](#)

Aisha  
Viewing as Student [Switch role](#)

[← All results](#)

## Science Basics — Quiz

You scored 20% — here's every question with the right answer.

1. Which gas do plants absorb during photosynthesis? X missed

Oxygen — your answer

Carbon dioxide ✓

Nitrogen

Hydrogen

Plants take in carbon dioxide.

2. What is the closest planet to the Sun? ✓ correct

Venus

Mercury ✓

Earth

Mars

Mercury is first from the Sun.

3. What organ pumps blood around the body? X missed

Lungs — your answer

Heart ✓

Brain

Liver

The heart pumps blood.

4. What force pulls objects toward Earth? X missed

Friction

Gravity ✓

Magnetism — your answer

Tension

Gravity pulls objects down.

5. How many legs does an insect have? X missed

4 — your answer

6 ✓

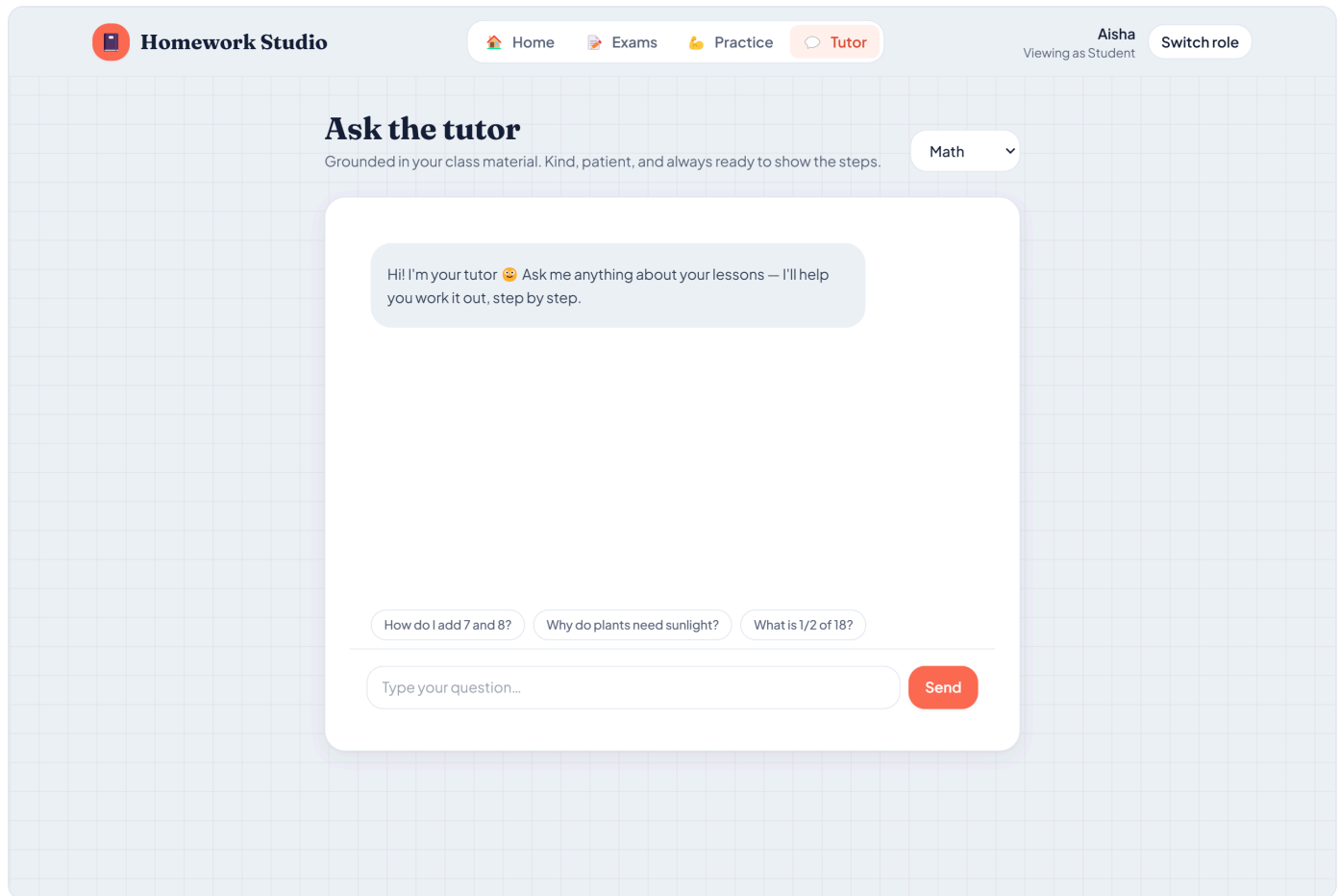
8

10

Insects have six legs.

## 5. Ask the tutor

Open **Tutor** to chat. It's patient, explains step by step, and answers from **your class's material** (the teacher's uploads), not the whole internet.



## Parent

**What you do:** see how each child is doing — grades *and* effort — get a tip on how to help, and open a printable report.

### 1. Your children

**My children** opens on a card per child showing not just grades but **engagement**: the overall average, a **day streak**, their **level**, and **badges earned** — with a "**needs a look**" flag for anyone dipping below 65%. Tap a card to see that child in detail. Below the cards you get:

-  **How to help** — a short, warm suggestion (AI-generated) built from your child's strongest subject and the area to grow.
- **This week's report** — written **automatically** by the system, no one pressing a button.
- **Progress over time** and **Compared to the class** (your child vs the class average per subject, with a marker for the class).
- **What's new** — a feed of reports, badges earned, new exams, and any at-risk alerts.
- The full **exam history**.

My children

How your children are doing — grades and effort — in plain language.

Progress report

Aisha Putri

Grade 4

on track

83% avg

5d

Lv 3

6

Math

100%

Science

67%

Rizky Putra

Grade 2

needs a look

53% avg

3d

Lv 1

3

Math

60%

Science

40%

Aisha's progress

How to help Aisha

Aisha clearly has a knack for numbers, so try baking together and let her do the measuring, doubling the recipe, and timing things — she'll be practicing math while discovering how science works in real life, all without it feeling like "studying."

This week's report

generated automatically

Aisha completed 5 exams this week with an overall average of 96%. Strongest in Math. Keep up the great effort! 🌟

83%

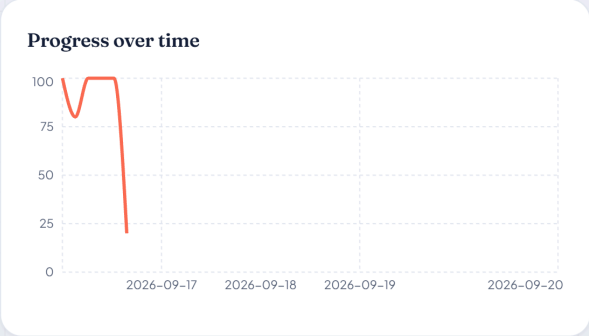
Overall average

6

Exams taken

5

Day streak



Compared to the class

The marker shows the class average.

Math

100% · ▲ 31 ahead

Science

67% · ▲ 5 ahead

What's new

Flagged Rizky Putra for extra support · 53%

2026-09-20

Report generated for Rizky Putra · 53%

2026-09-20

Report generated for Aisha Putri · 96%

2026-09-20

New exam published: Science Basics — Quiz

2026-09-20

New exam published: Math Basics — Quiz

2026-09-20

Aisha earned the "First steps" badge

Aisha earned the "Perfect!" badge

Rizky earned the "First steps" badge

Rizky earned the "Explorer" badge

Exam history

Science Basics — Quiz

Science · 2026-09-20

20% Done

|                       |                      |      |      |
|-----------------------|----------------------|------|------|
| Math Basics — Quiz    | Math · 2026-09-20    | 100% | Done |
| Science Basics — Quiz | Science · 2026-09-19 | 100% | Done |
| Math Basics — Quiz    | Math · 2026-09-18    | 100% | Done |
| Science Basics — Quiz | Science · 2026-09-17 | 80%  | Done |
| Math Basics — Quiz    | Math · 2026-09-16    | 100% | Done |

2. The printable report

Press **Progress report** for a clean, one-page report — a warm summary, headline numbers, strength by subject, and the full exam history. Use your browser's **Save as PDF** (the  button) to keep or share it.

 **Homework Studio**

 My children

Bu Sari (Parent)  
Viewing as Parent

Switch role

[← Back](#)

Save as PDF

 **Homework Studio**

Progress report · September 21, 2026

**Aisha Putri**  
Grade 4

Aisha has completed **6** exams with an overall average of **83%**. Strongest in Math (100%). A good area to keep practising is Science. Keep up the great effort! 🌟

83%  
Overall average

6  
Exams taken

**Math**  
Top subject

**By subject**

Math100%

Science67%

**Exam history**

| Date       | Title                 | Subject | Score |
|------------|-----------------------|---------|-------|
| 2026-09-16 | Math Basics — Quiz    | Math    | 100%  |
| 2026-09-17 | Science Basics — Quiz | Science | 80%   |
| 2026-09-18 | Math Basics — Quiz    | Math    | 100%  |
| 2026-09-19 | Science Basics — Quiz | Science | 100%  |
| 2026-09-20 | Math Basics — Quiz    | Math    | 100%  |
| 2026-09-20 | Science Basics — Quiz | Science | 20%   |

Generated by Homework Studio. Independent portfolio demo with synthetic data.

 **Admin**

**What you do:** monitor the whole school and watch the automation work.

## 1. School overview

**Overview** is the school at a glance: totals, the class average, and **mastery bands** (mastered / on track / needs support). The **Needs attention** panel surfaces students below 65% first. Below are top performers, strength by subject, the score distribution, and the full roster — each with a plain-language note.

## School overview

Everyone's progress in one place — with the students who need attention surfaced first.

Automation



6

Students



1

Teachers



1

Parents



65%

Class average



30

Exams taken



6

Auto-generated reports

### Mastery at a glance

How the class splits across mastery bands — so you can see the shape of the whole school, not just an average.



0

Mastered  
80%+

2

On track  
65-79%

4

Needs support  
below 65%

### Needs attention

Students below 65% — worth a check-in before they fall behind.

|                   |     |
|-------------------|-----|
| Rizky Putra       | 53% |
| Grade 2 - 3 exams |     |
| Budi Santoso      | 60% |
| Grade 4 - 5 exams |     |
| Dewi Lestari      | 60% |
| Grade 5 - 5 exams |     |
| Eka Pratama       | 64% |
| Grade 5 - 5 exams |     |

### Top performers

Recognise the students doing brilliantly this period.

1

Aisha Putri

74%

2

Chandra Wijaya

68%

3

Eka Pratama

64%

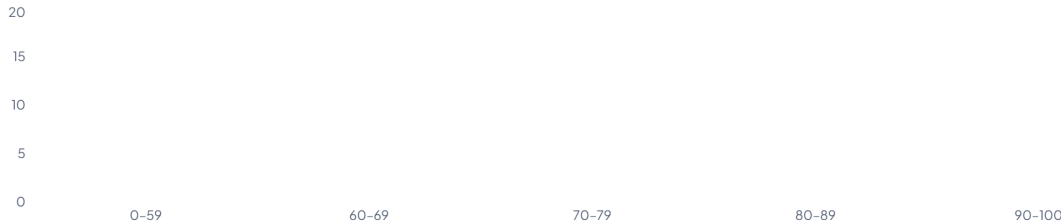
### Average by subject

Which subjects the school is strong in — and which need focus.



### Score distribution

How many exam attempts land in each score band — a healthy class leans right (green).



### All students

The full roster — sorted by average.

| Student        | Grade   | Average | Exams | Last activity |
|----------------|---------|---------|-------|---------------|
| Aisha Putri    | Grade 4 | 74%     | 7     | 2026-09-20    |
| Chandra Wijaya | Grade 4 | 68%     | 5     | 2026-09-20    |
| Eka Pratama    | Grade 5 | 64%     | 5     | 2026-09-20    |
| Budi Santoso   | Grade 4 | 60%     | 5     | 2026-09-20    |
| Dewi Lestari   | Grade 5 | 60%     | 5     | 2026-09-20    |
| Rizky Putra    | Grade 2 | 53%     | 3     | 2026-09-20    |

2. Insights

**Insights** goes beyond grades. **Engagement** shows how much the school is actually doing — students active today, average streak, total XP, badges earned, and the **streak leaders**. **Trend (last 14 days)** charts the school average and daily activity. **Teaching & exams** is the content pipeline — materials in, AI-generated exams out — with **per-exam performance** (takers, average, and the **hardest question**).

Homework Studio

OverviewInsightsAutomationMaterialsExams

Admin  
Viewing as Admin

Switch role

### Insights

Engagement, momentum, and the teaching pipeline — the whole school at a glance.

#### Engagement

How much the school is actually doing — effort, not just grades (from XP, streaks and badges).

**6**  
Active today

**4.7d**  
Avg streak

**1,022**  
Total XP

**14**  
Badges earned

#### Streak leaders

|         |    |      |        |
|---------|----|------|--------|
| Aisha   | 5d | Lv 3 | 278 XP |
| Chandra | 5d | Lv 2 | 168 XP |
| Eka     | 5d | Lv 2 | 164 XP |
| 4 Dewi  | 5d | Lv 2 | 160 XP |
| 5 Budi  | 5d | Lv 2 | 160 XP |

#### Trend (last 14 days)

The school's average exam score (line) and how many exams were taken each day (bars).

| Date  | Avg Score (%) | Exams Taken |
|-------|---------------|-------------|
| 09-07 | —             | —           |
| 09-08 | —             | —           |
| 09-09 | —             | —           |
| 09-10 | —             | —           |
| 09-11 | —             | —           |
| 09-12 | —             | —           |
| 09-13 | —             | —           |
| 09-14 | —             | —           |
| 09-15 | —             | —           |
| 09-16 | 72            | 5           |
| 09-17 | 68            | 5           |
| 09-18 | 70            | 6           |
| 09-19 | 68            | 6           |
| 09-20 | 55            | 8           |

#### Teaching & exams

The content pipeline — material in, AI-generated exams out — and how each exam is performing.

**3**  
Materials

**2**  
Published exams

**1**  
Drafts / in review

**1**  
Questions to review

#### Per-exam performance

| Exam                         | Status       | Takers | Average                    | Hardest question                                |
|------------------------------|--------------|--------|----------------------------|-------------------------------------------------|
| Science Basics — Quiz        | Published    | 13     | <div><div></div></div> 58% | How many legs does an insect have? · 8% correct |
| Math Basics — Quiz           | Published    | 17     | <div><div></div></div> 69% | 36 ÷ 6 = ? · 18% correct                        |
| Fractions for Grade 4 — Exam | Needs review | 0      | —                          | —                                               |

Homework Studio — independent portfolio demo

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### 3. Automation

**Automation** is where the background work shows up: the scheduler's status ("running · every 6h · **0 hands on it**"), how many reports it wrote and how many students it **auto-flagged**, which **AI models** are in use, and a **live activity feed**. Press ► **Run now** to trigger a run immediately.

Homework Studio

OverviewInsightsAutomationMaterialsExams

Admin  
Viewing as Admin

Switch role

Automation

What Homework Studio does on its own — reading, grading, reporting, and flagging without anyone pressing a button.

Run now

Weekly report scheduler — running

every 6h

6 reports generated

4 students auto-flagged

4m ago last run

0 hands on it

AI in use

Material reader: glm

Tutor: glm

Runs on a deterministic mock by default; real models activate with a key.

How the automation works

1 Reads & builds  
Uploaded material is read and turned into an exam.

2 Reports  
Each student's weekly report is written on a timer.

3 Flags  
Students who slip below 65% are flagged for support.

4 Delivers  
Parents just see it — no one pressed a button.

Recent activity

auto-refreshes

Flagged Rizky Putra for extra support · 53%

4m ago

Report generated for Rizky Putra · 53%

4m ago

Flagged Eka Pratama for extra support · 64%

4m ago

Report generated for Eka Pratama · 64%

4m ago

Flagged Dewi Lestari for extra support · 60%

4m ago

Report generated for Dewi Lestari · 60%

4m ago

Report generated for Chandra Wijaya · 68%

4m ago

Flagged Budi Santoso for extra support · 60%

4m ago

Report generated for Budi Santoso · 60%

4m ago

Report generated for Aisha Putri · 96%

4m ago

Admins can also open the teacher's **Materials** and **Exams** tools from the nav.

### Documents the system produces

Besides the exam, two other documents come out of one upload:

- **Teaching notes** — a lesson summary (key points + a worked example) generated from your material, shown on the material.
  - **The parent progress report** — a one-page, printable report per child (Save as PDF from the browser), refreshed automatically each week by the background scheduler.
- 

## Frequently asked

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**Do I need an API key to try it?** No. Everything runs on a free, built-in demo mode by default. Real AI models (reading PDFs/photos, richer exam generation, the tutor) turn on with a key.

**What can I upload, and how big?** One file per material: **PDF, PNG, JPG, TXT, or MD**, up to **25 MB**. ( `.docx` isn't supported yet.)

**Who grades the exams?** They're **auto-graded** the moment a student submits. The teacher's review is on the *questions* (before publishing), not on each student's answers.

**It works on my phone?** Yes — responsive to phone width, and installable (Add to Home Screen) as an app with an offline shell.

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See `TECHNICAL.md` for the architecture, data flow, and database design behind these screens.